Filter Design in the s-Domain

The most powerful application of the Laplace transform is the design of systems *directly* in the s-domain. This involves two steps: First, the s-domain is designed by specifying the number and location of the poles and zeros. This is a pure mathematical problem, with the goal of obtaining the best frequency response. In the second step, an electronic circuit is derived that provides this s-domain representation. This is something of an art, since there are many circuit configurations that have a given pole-zero diagram.

As previously mentioned, *step 4* of the Laplace transform method is very difficult if the system contains more than two poles or two zeros. A common solution is to implement multiple poles and zeros in *successive stages*. For example, a 6 pole filter is implemented as three successive stages, with each stage containing up to two poles and two zeros. Since each of these stages can be represented in the s-domain by a quadratic numerator divided by a quadratic denominator, this approach is called designing with **biquads**.

Figure 32-9 shows a common biquad circuit, the one used in the filter design method of Chapter 3. This is called the **Sallen-Key** circuit, after R.P. Sallen and E.L. Key, authors of a paper that described this technique in the mid 1950s. While there are several variations, the most common circuit uses two resistors of equal value, two capacitors of equal value, and an amplifier with an amplification of between 1 and 3. Although not available to Sallen and Key, the amplifiers can now be made with low-cost op amps with appropriate feedback resistors. Going through the four step circuit analysis procedure, the location of this circuit's two poles can be related to the component values:

EQUATION 32-4
Sallen-Key pole locations. These equations relate the pole position, ω and σ, to the amplifier gain, A, the resistor, R, and capacitor, C.

$$\sigma = \frac{A-3}{2RC}$$

$$\omega = \frac{\pm\sqrt{-A^2 + 6A - 5}}{2RC}$$

These equations show that the two poles always lie somewhere on a circle of radius: $1/RC$. The exact position along the circle depends on the gain of the amplifier. As shown in (a), an amplification of 1 places both of the poles on the real axis. The frequency response of this configuration is a low-pass filter with a relatively smooth transition between the passband and stopband. The -3dB (0.707) cutoff frequency of this circuit, denoted by ω_0, is where the circle intersects the imaginary axis, i.e., $\omega_0 = 1/RC$.